Assignment 2024
Deadline: Wednesday , 13th November at 4 pm

For this assignment you will assemble de-novo a set of short, paired-end reads from the *E.coli* O157:H7 strain, generated on the Illumina MiSeq sequencer, and assess whether adding long reads generated with Oxford Nanopore Technology (ONT) Minion devise improves the completeness and/or the accuracy of the alignment. You will perform the following sequence of tasks using the University of Glasgow's **Galaxy server**:

1. Assess the quality of the raw Illumina reads, using fastQC software.
2. Remove the adapters and trim the low-quality Illumina reads with Trim-Galore software (*use the same parameters as in our practical*)
3. Assess the quality of the trimmed Illumina reads, using fastQC software, and assess whether the quality of reads was improved.
4. Assemble trimmed Illumina reads into a set of contigs using SPAdes software.
5. Assess the completeness and accuracy of the assembly with Quast software.
6. Assemble trimmed Illumina reads AND long ONT reads together using SPAdes.
7. Assess the completeness and accuracy of the second assembly with Quast software.
8. Compare the key metrics and discuss pros and cons of each of the two assemblies.

You will prepare a written report of the above work in a form of a mini scientific paper. The structure of the report should be as follows:

1. Summary, brief description of what you have done and key results
2. Brief introduction about *E.coli* O157:H7 strain
3. Methods
    a. Short description of tools used in the assignment including FastQC, Trim-Galore, Spades and Quast
    b. Brief description of how the initial error-correction and post-assembly correction work.
4. Results.
    a. Provide briefly key FastQC metrics which you would use to demonstrate the quality improvement of Illumina reads after Trim-Galore preprocessing
    b. Record the initial number of Illumina reads, how many were removed due to quality-based trimming and how many were allowed into the assembly stage.
    c. Differentiate between two assemblies by comparison of key Quast generated metrics.
5. Discussion:
    a. If Trim-Galore improved reads quality, provide short explanation of why that happened.
    b. Given the comparison of assembly metrics in point 4.c. above, provide an argument as to which assembly you think is superior.
    c. Answer what you would do to improve the assemblies so that to obtain a single contig
6. Conclusions: formulate key conclusions
7. References: provide a list of key references you used.

The report should have no more than 2,000 words excluding figure legends and references.

Data sets are on Galaxy server:
Shared Data/Data Libraries/MScBioinformatics

Illumina reads:
1. TUV_S7_L001_R1_001.fastq
2. TUV_S7_L001_R2_001.fastq

Nanopore long reads
3. TUV359.2d.fq

Reference sequence to be used in Quast
4. Escherichia_coli_o157_h7_str_edl933_gca_000732965.ASM73296v1.31.dna.genome.fa